

The Neurochemistry of an Acute Social-Threat Stress Response: From Interruption to Sigh of Relief

TL;DR

- The felt "surge" in this scenario is the **sympathetic-adrenomedullary (SAM) arm** firing: the amygdala drives near-instant neural discharge of epinephrine/norepinephrine (adrenal secretion is neurally triggered and effectively immediate), producing $\beta 1/\beta 2/\alpha 1$ -adrenergic cardiovascular and metabolic effects; because catecholamines clear with a circulating half-life of ~1–3 minutes and the stressor was decisively terminated (the hang-up), the arousal drops fast and a vagally-mediated parasympathetic rebound — signalled by the physiological sigh — resets autonomic state.
- The **catharsis interpretation is wrong**: rapid relief after an outburst is a textbook **negative reinforcer** (removal of an aversive stimulus) plus a **relief-as-**

reward dopaminergic/opioid signal, and the best evidence shows venting *feeds* rather than extinguishes aggression — Kjærvik & Bushman's 2024 meta-analysis (154 studies, 10,189 participants) found arousal-decreasing activities reduced anger/aggression at $g = -0.63$ [95% CI $-0.82, -0.43$], with Bushman concluding "there's not a shred of scientific evidence to support catharsis theory" — meaning this loop tends to escalate, not discharge.

- The mechanistically honest correction to the "5–10 second adrenaline surge" folk model: adrenal secretion is near-instant, but *measured* plasma epinephrine rises over ~1–10 minutes and clears over minutes; the very brief subjective surge reflects perceptual/interoceptive onset and the decisive-action termination, not literal catecholamine kinetics — and the slower HPA/cortisol arm (peak ~20–30 min) barely engages in a stressor resolved in seconds.

Key Findings

1. **Two-arm activation.** An interpersonal confrontation is detected by the amygdala as a social threat and triggers two parallel stress axes on different timescales: the fast neural SAM axis (catecholamines, seconds) and the slower

hormonal HPA axis (cortisol, peaking ~20–30 min).

2. **The surge = SAM/catecholamines.** Adrenal medullary chromaffin cells are modified sympathetic neurons; secretion into blood is neurally driven and near-instant, hitting $\beta 1$ (heart), $\beta 2$ (bronchi/skeletal-muscle vasculature), and $\alpha 1$ (vasoconstriction) receptors. Psychological/emotional stressors preferentially raise epinephrine (Dimsdale & Moss 1980, *JAMA*: "During public speaking, epinephrine levels increase twofold, whereas during physical exercise, norepinephrine levels increase threefold ... psychological stress induces primarily an adrenal response").
3. **Rapid drop = kinetics + resolution.** Circulating epinephrine has a half-life of roughly 1–3 minutes (Eisenhofer 2001: catecholamine circulatory half-life <2 min), [ScienceDirect](#) cleared by uptake transporters plus COMT and MAO. A decisively *terminated* stressor stops further secretion, so levels fall quickly.
4. **Decisive action terminates the response.** The outburst + hang-up is a completed motor act that ends the aversive stimulus; this removes the threat input to the amygdala/PVN, permitting parasympathetic rebound. Contrast:

unresolved/helpless conflict sustains HPA drive and cortisol.

5. **The sigh is a real autonomic resetter.** Feldman & Krasnow (2016, *Nature*) mapped a brainstem peptidergic sigh circuit; Vlemincx and colleagues show sighs reset respiratory variability and produce relief; the long exhale increases vagal (parasympathetic) tone and slows the heart.
6. **Relief is reinforcing.** The removal of the aversive interruption negatively reinforces the outburst, and relief/termination of an aversive state activates mesolimbic dopamine (relief learning) and opioid signalling — a reward-learning loop that can condition and escalate anger over time. Catharsis is a myth.

Details

Step 0 — The trigger: being talked over as a social-threat signal

Repeated interruption is not a trivial annoyance to the nervous system; it functions as a **social-dominance / status-threat signal**. Classic conversation-analysis work (Zimmerman & West) tied interruption to social dominance and to who is expected to "fall silent," and subsequent research on power displays has associated interruption with status and dominance

assertion. Being repeatedly talked over — denied the conversational floor, unable to complete a thought — is registered as a loss of agency and a challenge to standing.

Neurally, threat appraisal is anchored in the **amygdala**, which is robustly implicated in processing social threat (rejection, exclusion, evaluation). The basolateral amygdala receives and processes the threat signal and passes it to the central (CeA) and medial (MeA) nuclei; the medial amygdala is specifically responsive to psychological (as opposed to systemic/inflammatory) stressors. [ScienceDirect](#) The amygdala then drives two outputs: (a) a fast projection (via hypothalamus and brainstem) that activates the sympathetic nervous system and adrenal medulla, and (b) a route to the hypothalamic paraventricular nucleus (PVN) that initiates the slow hormonal cascade. Crucially, prefrontal regions (especially right ventrolateral PFC) normally exert top-down inhibitory control over the amygdala; social rejection has been shown to degrade this regulatory capacity, biasing toward disinhibited, impulsive responses — i.e., the "blow-up."

Mounting frustration across repeated interruptions reflects successive amygdala activations without resolution, with catecholamines and arousal ratcheting up while prefrontal restraint is

progressively taxed — until the impulse breaks through as an outburst.

Step 1 — The fast neural arm: the SAM (sympathoadrenal) axis

The **sympathetic-adrenomedullary (SAM) system** is the body's rapid-response unit. On threat detection, the sympathetic branch of the autonomic nervous system fires in two ways:

- **Sympatho-neural:** postganglionic sympathetic fibers release **norepinephrine** directly onto target organs (heart, vasculature, lungs, gut).
- **Sympatho-adrenomedullary:** preganglionic splanchnic-nerve fibers synapse directly on **chromaffin cells of the adrenal medulla**, triggering release of **epinephrine (~80%) and norepinephrine (~20%)** into the bloodstream.

Because chromaffin cells are effectively modified sympathetic neurons, "catecholamine release is very quick and effects can be seen in less than a second" [ScienceDirect](#) at the secretory level — the adrenal medulla behaves like a sympathetic projection that discharges hormone into blood rather than into a synaptic cleft. [Springer](#)

Receptor targets and downstream effects:

- **$\beta 1$ (heart):** increased heart rate (positive chronotropy), contractility (positive inotropy), conduction velocity (positive dromotropy). Mechanism: $G_s \rightarrow$ adenylyl cyclase $\rightarrow$ cAMP $\rightarrow$ PKA $\rightarrow$ increased Ca^{2+} influx. [NCBI](#)
- **$\beta 2$ (bronchial and skeletal-muscle vascular smooth muscle):** bronchodilation and vasodilation to skeletal muscle — more air, more blood to muscles for action.
- **$\alpha 1$ (most vascular beds — skin, gut, kidney):** vasoconstriction, raising blood pressure and shunting blood to muscle and brain.
- **Metabolic:** glycogenolysis and lipolysis mobilize glucose and free fatty acids for fuel.

The subjective correlates — pounding/racing heart, chest tightness, flushed heat, trembling, heightened alertness — are the felt signature of this catecholamine action, and it is this that the person describes as the "adrenaline surge."

How fast, really? Here is an important accuracy correction. The mechanistic claim that adrenal effects appear "within a second" is true at the level of *secretion* (neural triggering of chromaffin cells) and of the *sympathetic-neural* effects on the heart and vessels. But human studies that actually *measured plasma catecholamines* show significant rises over the first **~1–10 minutes** of a stressor, because of

circulatory mixing/transit delay and sampling intervals. Dimsdale & Moss (1980, *JAMA* 243:340–342) documented epinephrine roughly doubling during public speaking; Oleshansky & Meyerhoff (1992, *Stress Medicine*) found EPI +82% and NE +67% within the first 10 minutes of a stressful interview. The fastest-resolving human method – continuous blood withdrawal (Ward et al. 1983, *Psychosomatic Medicine*) [PubMed Lippincott Williams & Wilkins](#) – pushes detection toward tens of seconds. So the reported **"5–10 second surge" is best understood as the interoceptive/perceptual onset** of catecholamine action (genuinely rapid because sympathetic *neural* effects on the heart and vasculature are near-instant), not as the full time-course of circulating adrenaline, which peaks and clears over minutes. Emotional/psychological stress is specifically an epinephrine-dominant (adrenal) response, whereas physical exertion is more norepinephrine-dominant (sympathetic-neural) [Semantic Scholar](#) – Dimsdale & Moss's key dissociation.

Step 2 – The slow hormonal arm: the HPA axis (and why it barely matters here)

In parallel, the amygdala (via CeA/MeA → bed nucleus of the stria terminalis → **PVN**, largely by disinhibition of GABAergic tone) drives the **hypothalamic-pituitary-adrenal (HPA) axis**:

- PVN **CRH** neurons release corticotropin-releasing hormone into the hypophyseal portal system;
- CRH stimulates the anterior pituitary to release **ACTH**;
- ACTH stimulates the adrenal **cortex** (zona fasciculata) to synthesize and release **cortisol**.

[Wiley Online Library](#)

Timescale: cortisol rises gradually, peaking **~20–30 minutes after stressor onset** and returning to baseline over the following **~60–90 minutes** [bioRxiv PubMed](#) (Kirschbaum & Hellhammer; "users guide to HPA axis research," Spencer & Deak 2017). Cortisol's functions are to sustain elevated blood glucose (gluconeogenesis), modulate immune/inflammatory tone, act on brain glucocorticoid and mineralocorticoid receptors to shape memory and feedback, and – via negative feedback – eventually shut the axis down.

The key clinical point for this scenario: a confrontation that is *resolved in seconds* engages the SAM axis fully but gives the HPA axis almost no time to matter – the surge is over long before cortisol would peak. The HPA/cortisol arm is the axis of **prolonged or unresolved** stress (rumination, helplessness, ongoing conflict). This is precisely why *resolving* the stressor by decisive action produces a

clean, brief arousal rather than the lingering, cortisol-soaked malaise of an argument left hanging.

Step 3 – Catecholamine kinetics: why the surge is short

Catecholamines are cleared from blood extraordinarily fast. The circulating **half-life of epinephrine is ~1–3 minutes** [ScienceDirect](#) (commonly cited as ~2 min); some physiology sources quote a biological half-life of 10–100 seconds for catecholamines generally, [ScienceDirect](#) and Eisenhofer (2001, *Pharmacol Ther* 91:35–62) frames the catecholamine **circulatory half-life as <2 minutes**. Clearance occurs by:

- **Neuronal reuptake (uptake-1)** and **extraneuronal uptake (uptake-2)** membrane transporters – the dominant, very rapid route;
- Enzymatic metabolism by **catechol-O-methyltransferase (COMT)** → metanephrine/normetanephrine, and **monoamine oxidase (MAO)** → deaminated metabolites, converging on **vanillylmandelic acid (VMA)**, [ScienceDirect](#) with conjugation and renal excretion.

The combination of two facts explains the felt 5–10-second surge → rapid drop:

1. **The stressor was decisively terminated** (hang-up), so the amygdala stops driving sympathetic outflow and adrenal secretion **stops** — no new catecholamine is entering the blood.
2. **Rapid clearance** (half-life of minutes, transporter uptake in seconds) means existing catecholamines are mopped up quickly.

With no ongoing secretion and fast clearance, arousal collapses. (Individual COMT genotype matters: the Val158Met/rs4680 polymorphism modulates clearance speed — Val/Val "warriors" clear catecholamines faster; Met/Met clear slower and may feel prolonged arousal.) [Latrinawaldenexamsolutions](#)

Step 4 — Decisive action / motor discharge as a terminator

A central organizing principle: the acute stress response is a **preparation for action**, and completing an action that *ends the stressor* removes the input that sustains the response. The outburst + hang-up accomplishes exactly this — it terminates the aversive interruption and restores agency/control in one decisive motor act. Once the threat signal to the amygdala ceases, sympathetic drive withdraws and the parasympathetic system can rebound.

This is the sharp contrast the scenario illustrates:

- **Stress resolved by action** → sympathetic drive stops → fast catecholamine clearance → parasympathetic rebound → calm. (Minimal HPA involvement.)
- **Stress that persists without resolution** (helplessness, being unable to end the interruptions, unresolved conflict, rumination) → continued amygdala/PVN drive → sustained HPA activation and elevated cortisol → the physiologically costly, prolonged state associated with allostatic load. Note the important nuance: perceived *controllability* is itself a major modulator of the response — animal and human work shows controllable/escapable stressors produce smaller, shorter neuroendocrine responses than uncontrollable ones.

A caution against over-reading this: "discharge" here means *the stressor ended*, not that aggressive expression drained a hydraulic reservoir of anger. The relief comes from **threat termination**, which is a very different mechanism from catharsis (see Step 6).

Step 5 — Parasympathetic rebound and the physiological sigh

Parasympathetic reactivation. After sympathetic withdrawal, the **parasympathetic ("rest and digest") system**, via the **vagus nerve**, reasserts control. The vagus (through the nucleus ambiguus and dorsal

motor nucleus) slows the sinoatrial node, dropping heart rate and blood pressure. [uspto](#) Empirically, **"vagal rebound"** — a sharp rise in heart-rate variability in the first minute of recovery — is a documented marker of healthy recovery from acute psychological stress (Mezzacappa et al. 2001, *Psychosomatic Medicine*): cardiovascular recovery is associated with increased vagal modulation even while some residual sympathetic activation lingers, [PubMed](#) and blunted vagal rebound predicts cardiovascular risk. After a stressor passes, parasympathetic activity typically overshoots past baseline before settling.

[ResearchGate](#)

The physiological sigh. A sigh is a **double inhalation (a second "top-up" breath stacked on the first) followed by a long, extended exhalation.** Feldman (UCLA) and Krasnow (Stanford) identified the generating circuit (Li et al. 2016, *Nature* 530:293–297, "The peptidergic control circuit for sighing"): ~200 neurons expressing the bombesin-related peptides **neuromedin B (NMB)** or **gastrin-releasing peptide (GRP)** in the retrotrapezoid nucleus/parafacial respiratory group (RTN/pFRG) project to ~200 receptor-expressing neurons in the **preBötzinger complex** (the breathing rhythm generator) to convert a normal breath into a sigh — defined in that paper as "long, deep breaths expressing sadness, relief or exhaustion." Silencing both peptides in mice

abolished sighing. Humans sigh spontaneously roughly every 5 minutes (~12/hour) to reinflate collapsed alveoli.

Why a sigh accompanies relief — and how it resets autonomic state:

- **Alveolar reinflation and gas exchange:** the double-inhale pops open collapsed alveoli, correcting the disrupted O_2/CO_2 balance caused by the rapid, shallow breathing of stress.
- **Respiratory-variability reset:** Elke Vlemincx and colleagues developed the "**resetter**" model — stress and negative affect increase *random* respiratory variability; a sigh restores *structured/correlated* variability [ScienceDirect](#) and is followed by measured relief and reduced muscle tension (Vlemincx et al. 2010, 2016, "A sigh of relief or a sigh to relieve"). Sigh rate rises specifically at **emotional transitions** and during relief, [ScienceDirect](#) which is why a sigh spontaneously accompanies the shift from threat to safety.
- **Vagal braking via the long exhale:** the prolonged exhalation is the key autonomic lever. During inhalation, pulmonary stretch receptors (via the vagus, the **Hering-Breuer reflex**) transiently inhibit cardiac vagal neurons, speeding the heart; during exhalation, this inhibition is *released*, so vagal

tone rises and heart rate falls (respiratory sinus arrhythmia). [uspto](#) An extended exhale therefore maximizes parasympathetic (vagal) braking. This is the mechanism behind the Stanford "cyclic sighing" study (Balban et al. 2023, *Cell Reports Medicine* 4(1):100895; RCT NCT05304000, ~108–111 participants): "Breathwork, especially the exhale-focused Cyclic Sighing produces greater improvement in mood ($p < 0.05$) and reduction in respiratory rate ($p < 0.05$) compared to Mindfulness Meditation" over one month — though notably, that study found **no significant change in HRV or resting heart rate** across groups, a limitation worth flagging.

Endorphin / tension-release component. Acute stress and intense/anaerobic exertion measurably raise plasma **β -endorphin** (co-released with ACTH from the pituitary POMC precursor): Schedlowski et al. (1995, *Psychoneuroendocrinology* 20:103–110) found a transient significant β -endorphin rise immediately after a first parachute jump; stress can also produce naloxone-reversible (opioid-mediated) analgesia in humans (Willer, Dehen & Cambier 1981, *Science* 212:689–691). This plausibly contributes to the "clean," de-tensed post-outburst feeling. **Important caveat:** peripheral β -endorphin crosses the blood–brain barrier poorly, so attributing the *subjective* clean feeling specifically to circulating β -endorphin is

mechanistically contested — the mood effect more likely reflects central opioid/endocannabinoid systems plus the sheer removal of threat and the vagal reset. Treat the endorphin link as a plausible associated contributor, not an established cause.

Step 6 — The reinforcement / conditioning angle (why this loop is dangerous)

This is where the mechanism becomes clinically important, because the *sequence* — aversive interruption → outburst → threat removed → intense relief — is a near-perfect conditioning paradigm.

Negative reinforcement. In operant terms, **negative reinforcement** *increases* a behavior by *removing* an aversive stimulus. Here the aversive stimulus is the interruption/conflict; the behavior is the outburst + hang-up; the consequence is removal of the aversive stimulus plus a rush of relief. That relief **reinforces the outburst**, making it more likely next time. Note this is *reinforcement*, not punishment — the "negative" means subtraction of something aversive, and the behavior is strengthened.

Operant conditioning of anger/aggression. Anger and aggression are learnable instrumental behaviors (Bandura). When aggression "works" — ends the conflict, restores control, produces relief — it is **instrumentally reinforced**. Bushman's program of

research reframes much everyday aggression as **affect regulation**: people aggress to improve how they feel (Bushman, Baumeister & Phillips 2001, "Do people aggress to improve their mood?"). If the outburst reliably terminates the aversive state, the brain tags it as an effective mood-repair strategy.

Relief as reward – the neurochemistry. The termination of an aversive state is itself a reward signal:

- **Dopamine (relief learning):** midbrain dopamine neurons fire at the *cessation* of aversive stimuli. [ScienceDirect](#) A specific projection from the posteromedial ventral tegmental area (pmVTA) to the nucleus accumbens shell mediates **relief learning** (*Neuropsychopharmacology* 2018); dopamine is released in the NAc within milliseconds of aversive-stimulus termination (Navratilova et al. 2015, *Trends in Neurosciences*, "Brain circuits encoding reward from pain relief"). [PubMed](#)
- **Opioids:** relief of an aversive/painful state requires opioid signalling in the anterior cingulate cortex, functionally linked to accumbens dopamine release, to produce the rewarding, pleasurable quality of relief. [PubMed](#)

So the calm-and-sigh that follows the outburst is not neutral – it is a **dopaminergic/opioidergic reward**

stamped onto the immediately preceding behavior.
This is the reward-learning loop: **trigger → outburst → threat removed → dopamine/opioid relief signal → strengthened outburst behavior.**

The catharsis myth. The folk "hydraulic" model — that venting anger discharges it like releasing steam — is **empirically false**. Bushman (2002, *Personality and Social Psychology Bulletin* 28(6):724–731, "Does venting anger feed or extinguish the flame?") found that participants who vented (hit a punching bag while thinking of their provoker) were *angrier and more aggressive* afterward than distraction or do-nothing controls; in his words: "Rumination increased rather than decreased anger and aggression. Doing nothing at all was more effective than venting anger. These results directly contradict catharsis theory." The 2024 meta-analysis (Kjærvik & Bushman, *Clinical Psychology Review*, 154 studies, 10,189 participants) confirmed that **arousal-decreasing activities** (deep breathing, relaxation, mindfulness, meditation, slow-flow yoga, progressive muscle relaxation, diaphragmatic breathing, timeout) reduce anger — pooled effect $g = -0.63$ [95% CI $-0.82, -0.43$] — while **arousal-increasing activities** (venting, hitting bags, even jogging) do not, and can make it worse. Bushman's summation: "there's not a shred of scientific evidence to support catharsis theory," and

"any good feeling we get from venting actually reinforces aggression."

The apparent paradox — the person *feels* better after the outburst — is fully explained by the mechanisms above: the relief is real (threat was removed; parasympathetic rebound; possible opioid/dopamine reward), but that very relief is the *reinforcer* that entrenches the aggressive response. Feeling better ≠ becoming less aggressive. Catharsis confuses the immediate hedonic relief with a reduction in the disposition to aggress; the data show the disposition is *strengthened*.

Clinical escalation and what breaks the loop.

- **Escalation:** each cycle strengthens the association, can generalize to lower provocation thresholds, and — via stress sensitization — can lower the bar for future reactivity. Over time the outburst can become the *default* down-regulation strategy for that trigger.
- **What breaks it:** (1) Insert an **arousal-reducing** strategy at the moment of provocation (the physiological sigh / long exhale, timeout) instead of the outburst — this achieves the autonomic reset *without* reinforcing aggression. (2) Strengthen **prefrontal top-down control** (the rVLPFC over amygdala/accumbens) via impulse-control and reappraisal training; CBT-style work on

the *meaning* attached to interruption reduces the amygdala threat appraisal itself. (3) Remove the reinforcement contingency – e.g., assertive, non-aggressive means of "reclaiming the floor" that still terminate the aversive stimulus, so relief is obtained without the aggressive act being the thing that produced it. (4) Address the trigger environment (reduce exposure to the specific aversive interruption pattern).

Step 7 – Individual differences and modulators

- **Baseline vagal tone / HRV:** higher resting vagally-mediated HRV is associated with better emotion regulation, impulse control, and stress recovery; lower resting HRV predicts greater difficulties in emotion regulation (especially impulse control) [nih](#) and is broadly (though not unanimously) linked to trait hostility/aggression. A person with low baseline vagal tone has both a weaker "brake" and a slower rebound. (Note: the literature is mixed – a recent meta-analysis found the HRV–reactive-aggression link is not robust, [Wiley Online Library](#) so treat vagal tone as a contributor, not a determinant.)
- **Trait aggression / hostility / anger-out:** individuals high in trait anger or "anger-out" tendencies show larger aggressive responses and

are more susceptible to the venting-reinforcement trap (Bushman's Study 2 samples).

- **Stress sensitization:** prior stress exposure can cross-sensitize the HPA axis and threat circuitry, lowering the threshold for future activation — especially after high-intensity emotional stressors. [Taylor & Francis Online](#)
- **COMT / MAO genotype:** Val158Met (rs4680) and MAOA-uVNTR polymorphisms modulate catecholamine clearance and thus the *duration* of the surge.
- **Testosterone / dominance orientation:** testosterone biases the amygdala toward *threat approach* (rather than avoidance), which in a dominance-challenge like being talked-over may tilt the response toward confrontation.
- **Low tolerance for a specific trigger:** a person for whom being silenced/interrupted is a particularly loaded aversive cue (learning history, prior experiences of not being heard) will appraise it as a stronger threat and respond more intensely — and will find its removal correspondingly more reinforcing.

Recommendations

Framing for the workshop/GitHub issue: Present the workflow as a mapped pipeline — *interruption*

(amygdala social-threat appraisal) → mounting frustration (accumulating SAM arousal, taxed PFC) → outburst + hang-up (decisive motor act terminating the stressor) → surge (catecholamine/SAM, felt onset near-instant, cleared in minutes) → calm + sigh (parasympathetic/vagal rebound, physiological-sigh reset) → **reinforcement trap** (relief negatively reinforces the outburst; dopamine/opioid relief-reward). Make the reinforcement node visually prominent because it is the counterintuitive, decision-relevant part.

Staged actionable guidance:

1. **In the moment (acute down-regulation without reinforcing aggression):** deploy a **physiological sigh** (double inhale, long exhale) or a few exhale-emphasized breaths, or take a **timeout**, *before* reacting. This gets the same autonomic reset the outburst produced — but without stamping in the aggression.
2. **Skill-building (weeks):** practice **cyclic sighing / longer-exhale breathing ~5 min/day** (Balban et al. 2023) to raise the resting parasympathetic set-point; add cognitive reappraisal of interruption ("this is their impulsivity/anxiety, not a verdict on my worth") to reduce the amygdala threat signal at the source.

3. **Structural (address the contingency):** develop assertive floor-reclaiming scripts that terminate the aversive interruption *non-aggressively*, and reduce exposure to the trigger where feasible.
4. **Escalate to professional support if:** outbursts are increasing in frequency/intensity, generalizing to new triggers, damaging relationships/work, or accompanied by loss of control — indicators the reinforcement loop is entrenching and warranting CBT / anger-regulation therapy.

Benchmarks that change the plan: If in-the-moment arousal-reduction reliably preempts outbursts and their frequency falls, self-management is working. If outbursts persist or escalate despite consistent practice — or if the relief is being actively *sought* (aggressing to feel better) — that signals the operant loop is dominant and clinical intervention (CBT for anger, addressing trauma/beliefs that fuel the threat appraisal) is indicated.

Caveats

- The "5–10 second surge" is a **subjective/interoceptive descriptor, not literal plasma-catecholamine kinetics**. Adrenal *secretion* and sympathetic *neural* cardiac effects are near-instant, but *measured* plasma epinephrine rises over ~1–10 minutes and clears over minutes

(half-life ~1–3 min). The felt brevity reflects fast neural onset + decisive stressor termination + rapid clearance, not a literal 5–10-second hormone spike.

- **"Discharge" is not catharsis.** The relief comes from *threat termination* and autonomic rebound, not from draining a reservoir of anger. The hydraulic/steam metaphor is empirically refuted (Bushman 2002; Kjærvik & Bushman 2024).
- **The endorphin contribution to the "clean feeling" is plausible but not established** — plasma β -endorphin rises with acute stress, but its poor BBB penetration means the subjective effect is more likely central (opioid/endocannabinoid) and/or simply the relief of threat removal plus vagal reset.
- **Individual-difference links are probabilistic, not deterministic** — the HRV–aggression association in particular is contested (a recent meta-analysis disagreed with the standard account). Treat vagal tone, genotype, and trait measures as modulators, not predictors of any single episode.
- **Some cited material is popular-science or secondary** (e.g., Huberman-popularized framing of sighs); the underlying primary sources (Li/Feldman/Krasnow 2016; Vlemincx; Balban 2023; Bushman) are peer-reviewed and are the load-bearing citations.

- This is a **generic mechanistic workflow example**, not clinical advice for any specific individual.